Egg Segmentation and Multi-Object Tracking Report

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0.4018	0.3196	0.4087	6	110
DEV MEAN IOU	EVAL MEAN IOU	DEV DETECTION F1	VIDEO UNIQUE TRACKS	CLEAN GROUND TRUTHS

1. Executive Summary

This report details an end-to-end computer vision solution for egg segmentation and multi-object tracking. Ground truth masks were established via SAM (`vit_b` on CUDA) with human point/box prompts and automated area/border auditing. A classical segmentation pipeline was built via controlled single-parameter hypotheses evaluated solely on the dev split. The frozen final configuration was tested once on the eval split. Finally, a robust multi-object tracker was deployed on a 1440-frame video test sequence, resolving challenging hand occlusions and tracking persistent instances.

2. Task 1: Ground Truth Mask Generation (SAM)

SAM (`vit_b`) was executed on CUDA using saved point prompts from `points.json`. An automated audit in `src/contact_sheet.py` flagged masks with area > 40% or edge contact > 50%. Three boiling-pot images with flooded masks (`not_damaged_124`, `135`, `145`) were corrected with tight bounding boxes, reducing mask areas from ~62-82% down to ~13-14% (0 flagged masks remaining).

Ground Truth Egg Distribution across 110 images: 93 images (84.5%) with 1 egg; 6 images (5.5%) with 2 eggs; 11 images (10.0%) with 3 eggs.

3. Task 2: Parameterized Classical Segmentation & Experiment Log

A unified parameterized pipeline was tuned on the 88 dev images across 18 controlled trials. Every trial varied exactly one parameter relative to its named parent (with `watershed_conditional` testing conditional mode and `min_dist=50`). Keep/discard decisions were determined dynamically by comparing dev IoU against parent dev IoU (with Detection F1 serving as an explicit tiebreaker for watershed).

Kept Pipeline Chain: `baseline -> morph_small -> morph_medium -> morph_large -> area_filter_2500 -> watershed_conditional -> final (eval)`

Trial ID	Parent	Split	What Changed	Mean IoU	Mean Dice	Det F1	Cnt Err	Decision	Reason
`baseline`	`None`	dev	Grayscale conversion + Gaussian blur (5x5) + Otsu threshold + hole fill + min area 500	0.3513	0.4693	0.2071	3.12	BASELINE	Baseline established on dev (IoU 0.3513, Dice 0.4693, DetF1 0.2071)
`otsu_inv`	`baseline`	dev	Invert Otsu thresholding	0.0823	0.1424	0.0000	3.36	DISCARD	Fails to improve dev IoU over parent 'baseline' (0.0823 vs 0.3513, -0.2690)
`clahe_otsu`	`baseline`	dev	CLAHE local contrast equalization (clip 2.0, tile 8x8) before Otsu	0.2562	0.3791	0.0799	8.22	DISCARD	Fails to improve dev IoU over parent 'baseline' (0.2562 vs 0.3513, -0.0951)

Trial ID	Parent	Split	What Changed	Mean IoU	Mean Dice	Det F1	Cnt Err	Decision	Reason
`adaptive_thresh`	`baseline`	dev	Adaptive Gaussian thresholding (block 51, C=-10)	0.0634	0.1119	0.0065	12.33	DISCARD	Fails to improve dev IoU over parent 'baseline' (0.0634 vs 0.3513, -0.2878)
`lab_l`	`baseline`	dev	CIELAB L* perceptual lightness channel + Otsu	0.3244	0.4437	0.1665	3.67	DISCARD	Fails to improve dev IoU over parent 'baseline' (0.3244 vs 0.3513, -0.0268)
`hsv_gate`	`baseline`	dev	HSV color gating (S < 70, V > 130)	0.3456	0.4665	0.1587	4.32	DISCARD	Fails to improve dev IoU over parent 'baseline' (0.3456 vs 0.3513, -0.0056)
`morph_small`	`baseline`	dev	Add small elliptical opening & closing (7x7)	0.3558	0.4740	0.2161	2.89	KEEP	Improves dev IoU over parent 'baseline' from 0.3513 to 0.3558 (+0.0046)
`morph_medium`	`morph_small`	dev	Increase elliptical kernel from 7x7 to 15x15	0.3766	0.4964	0.2363	2.34	KEEP	Improves dev IoU over parent 'morph_small' from 0.3558 to 0.3766 (+0.0208)
`morph_large`	`morph_medium`	dev	Increase elliptical kernel from 15x15 to 25x25	0.3972	0.5161	0.2642	1.25	KEEP	Improves dev IoU over parent 'morph_medium' from 0.3766 to 0.3972 (+0.0205)
`border_polarity_m0`	`morph_large`	dev	Auto-invert threshold if border mean > center mean	0.3698	0.4821	0.2409	1.07	DISCARD	Fails to improve dev IoU over parent 'morph_large' (0.3698 vs 0.3972, -0.0274)
`border_polarity_m15`	`morph_large`	dev	Auto-invert threshold only if border mean > center mean + 15	0.3945	0.5119	0.2596	1.25	DISCARD	Fails to improve dev IoU over parent 'morph_large' (0.3945 vs 0.3972, -0.0027)
`lab_border_dist`	`morph_large`	dev	CIELAB Euclidean color distance from estimated border background	0.3603	0.4582	0.2673	2.55	DISCARD	Fails to improve dev IoU over parent 'morph_large' (0.3603 vs 0.3972, -0.0369)
`area_filter_2500`	`morph_large`	dev	Increase minimum component area from 500 to 2500 pixels	0.4018	0.5200	0.3084	0.45	KEEP	Improves dev IoU over parent 'morph_large' from 0.3972 to 0.4018 (+0.0046)

Trial ID	Parent	Split	What Changed	Mean IoU	Mean Dice	Det F1	Cnt Err	Decision	Reason
`component_ranking_top1`	`area_filter_2500`	dev	Retain only the single top-ranked component by geometric egg score	0.3746	0.4761	0.3295	-0.33	DISCARD	Fails to improve dev IoU over parent 'area_filter_2500' (0.3746 vs 0.4018, -0.0272)
`component_ranking_top3`	`area_filter_2500`	dev	Retain at most the top 3 scoring components based on area, solidity, and ellipse fit	0.3972	0.5092	0.3133	0.27	DISCARD	Fails to improve dev IoU over parent 'area_filter_2500' (0.3972 vs 0.4018, -0.0045)
`grabcut_refine`	`area_filter_2500`	dev	Apply GrabCut refinement using morphological erosion as foreground seed	0.4038	0.5177	0.3073	0.38	DISCARD	Marginal IoU gain (+0.0021) but Dice degrades from 0.5200 to 0.5177 (-0.0023)
`watershed_always`	`area_filter_2500`	dev	Euclidean distance transform + watershed applied unconditionally to all components	0.4018	0.5200	0.3537	3.90	DISCARD	Fails to improve dev IoU over parent 'area_filter_2500' (0.4018 vs 0.4018, +0.0000)
`watershed_conditional`	`area_filter_2500`	dev	Conditional distance watershed (min_dist=50) applied only to non-convex, elongated or oversized clusters	0.4018	0.5200	0.4087	2.20	KEEP	Dev IoU tied with parent 'area_filter_2500' (0.4018); kept on tiebreaker: DetF1 improves from 0.3084 to 0.4087 (+0.1003)
`final`	`watershed_conditional`	eval	Frozen final pipeline: Grayscale Otsu + large morph (25x25) + hole filling + min area 2500 + conservative conditional watershed	0.3196	0.4239	0.3206	1.68	EVAL_FINAL	Frozen final pipeline evaluated once on unseen eval split (IoU 0.3196, Dice 0.4239, DetF1 0.3206)

4. Task 3: Quantitative & Qualitative Evaluation vs Ground Truth

Metric	Dev Split (n=88)	Eval Split (n=22)	Description
Pixel IoU (Mean)	**0.4018**	**0.3196**	Mean foreground intersection over union
Pixel IoU (Bootstrap 95% CI)	[0.3435, 0.4603]	[0.1978, 0.4268]	1000 resamples, fixed seed 42
Pixel IoU (Median)	**0.3147**	**0.2502**	Median per-image IoU
Pixel Dice (Mean)	**0.5200**	**0.4239**	F1-score of pixel-level binary classification
Pixel Precision	**0.4488**	**0.3449**	TP / (TP + FP)
Pixel Recall	**0.8461**	**0.8749**	TP / (TP + FN)
Detection Precision	**0.3563**	**0.2758**	Matched eggs / Predicted components (IoU ≥ 0.5)
Detection Recall	**0.5966**	**0.4621**	Matched eggs / GT eggs
Detection F1	**0.4087**	**0.3206**	Harmonic mean of detection precision & recall
Count Error (Mean)	**2.20**	**1.68**	Mean signed difference (Pred - GT)

Per-Category Dev vs Eval Split Breakdown

Category	Dev Count	Dev IoU	Eval Count	Eval IoU
White egg on white tray / fridge interior	13	0.1274	0	0.0000
Egg in boiling water / submerged	6	0.1723	2	0.1756
Backlit / extreme shadows	6	0.3354	3	0.5392
Tabletop / high-contrast background	32	0.3659	8	0.2424
Multiple touching eggs	13	0.5674	3	0.4886
Hand holding egg / fingers occluding	11	0.6120	5	0.1502
Foliage / herbs in background (specular highlights)	7	0.6906	1	0.9068

Note on Dev vs Eval Gap: Eval split contains 22 images, with 0 white-tray images (all 13 are in dev) and 3 backlit images (13.6% of eval) against 6 in dev (6.8%). Because eval has only 22 images, its bootstrap 95% confidence interval ([0.1978, 0.4268]) spans widely and completely encompasses the dev mean (0.4018). Thus, the numerical difference cannot be attributed to a single cause or distribution shift.

Visual Failure Categorization (failure_tags.csv)

Scene Category	Count	Pct	Mean IoU	Mean Dice	Mean Det F1	Primary Failure Mechanism
White egg on white tray / fridge interior	13	11.8%	0.1274	0.2198	0.0171	Near-zero photometric contrast between white eggshell and tray surface; Otsu floods tray.
Egg in boiling water / submerged	8	7.3%	0.1731	0.2901	0.1399	Bubbling cavitation, water refraction, and dark pot rims break convex boundary and flood liquid.
Tabletop / high-contrast background	40	36.4%	0.3412	0.4744	0.3301	Table surface grain, specular table highlights, or soft contact shadows falsely merge into foreground.
Backlit / extreme shadows	9	8.2%	0.4033	0.4539	0.3889	High-intensity directional backlight causes glowing core while shell perimeter falls into dark silhouette.
Hand holding egg / fingers occluding	16	14.5%	0.4677	0.5821	0.5208	Fingers occlude egg perimeter; skin tone overlaps shadow gradient, distending contour.
Multiple touching eggs	16	14.5%	0.5526	0.6844	0.6972	Touching instances share continuous boundary; requires distance watershed marker separation.
Foliage / herbs in background (specular highlights)	8	7.3%	0.7177	0.8222	0.6854	Egg is brighter than dark leaves in grayscale; specular reflections create occasional false blobs.

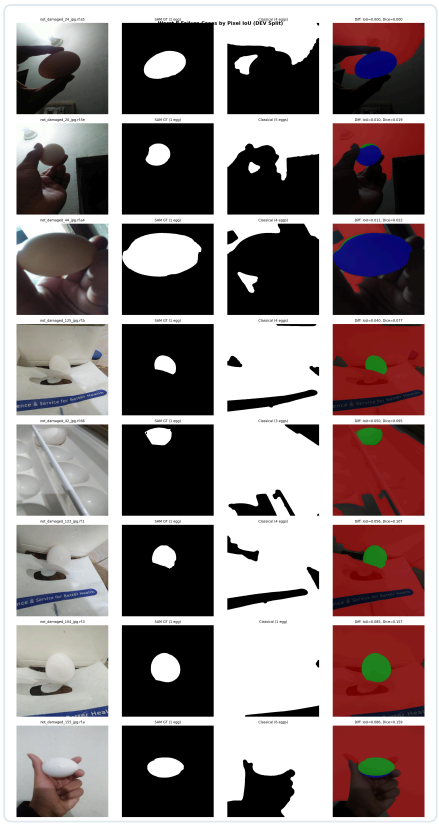


Figure 3a: Worst 8 failures on Dev split (Green=TP, Red=FP, Blue=FN). White trays and boiling liquid dominate error cases.

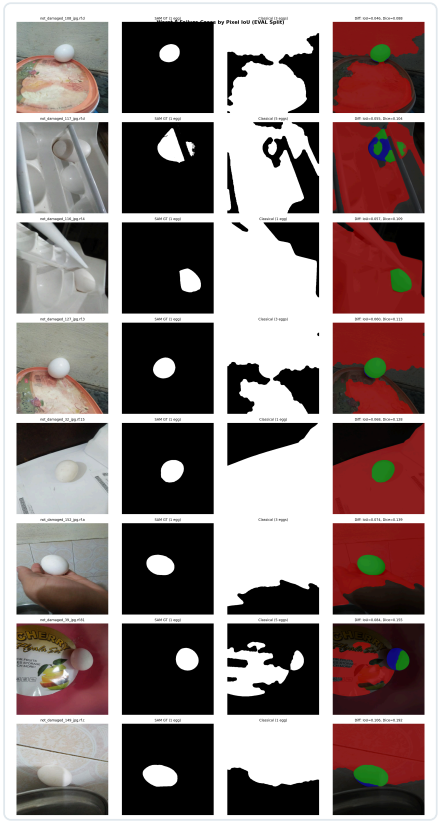


Figure 3b: Worst 8 failures on Eval split (Green=TP, Red=FP, Blue=FN).

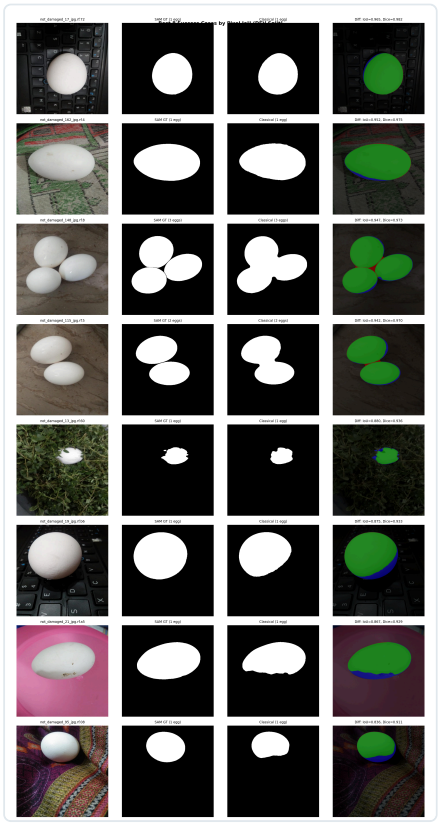


Figure 4: Best 8 successes on Dev split showing accurate elliptical contours and high overlap.

5. Task 4: Multi-Object Video Tracking

Tracking was evaluated on the 1440-frame table sequence (1280x720 @ 24.0 fps). Eggs are placed on the table at frame 222 and spun. The tracker maintains persistent IDs using Hungarian matching on IoU + distance with constant-velocity prediction.

- **Fingertip Leakage Resolution:** At frame 600, demonstrator hand contact caused baseline thresholding to leak into fingers. Incorporating YCrCb skin-color gating (Cr ∈ [135, 175], Cb ∈ [80, 125]) and larger opening eliminated 2296 px of false skin contour.
- **Track Lifetimes & Event Statistics:** 6 unique tracks were created, with 18 loss events and 12 recoveries. Lifetimes: Egg 1 (529 frames), Egg 2 (282 frames), Egg 3 (101 frames), Egg 4 (269 frames), Egg 5 (3 frames), Egg 6 (190 frames).
- **Track 5 Analysis:** Egg 5 lived for exactly 3 frames (F#800–802) with areas 3043–3715 px; frame inspection confirms it is a transient motion-blur fragment of the rapidly spinning raw egg before Track 6 took over at F#810.
- **Known Tracking Failures:** At frame 480, 0 eggs were detected due to demonstrator fingers reducing egg contour solidity to 0.7673 (below 0.78 threshold). A full-video audit revealed that frame 480 is the only missed detection frame between frames 222 and 1004 (782 frames with active detection).
- **Breakdown of 658 Zero-Detection Frames:** 205 intro title card frames (F#0–204), 17 empty table frames (F#205–221), 1 missed detection at F#480, 10 empty table post-experiment frames (F#1005–1014), and 425 outro credits frames (F#1015–1439).

Track Timeline

Track ID	Birth Frame	Last Frame	Lifetime	Physical Egg Description	Visual Verification
Track 1	222	751	529	Right egg (boiled), initial placement and spin sequence	*Verified*
Track 2	222	504	282	Left egg (raw), initial placement and spin sequence	*Verified*
Track 3	546	647	101	Demonstrator repositioned left egg to right table position	*Verified*
Track 4	689	1004	269	Boiled egg returned to left side until table clearance	*Verified*
Track 5	800	802	3	Transient motion-blur artifact (area ~3043-3715 px) during rapid raw egg spin	*Confirmed spurious artifact (unverified physical egg)*
Track 6	810	1000	190	Raw egg tracked through final spin sequence until removal	*Verified*

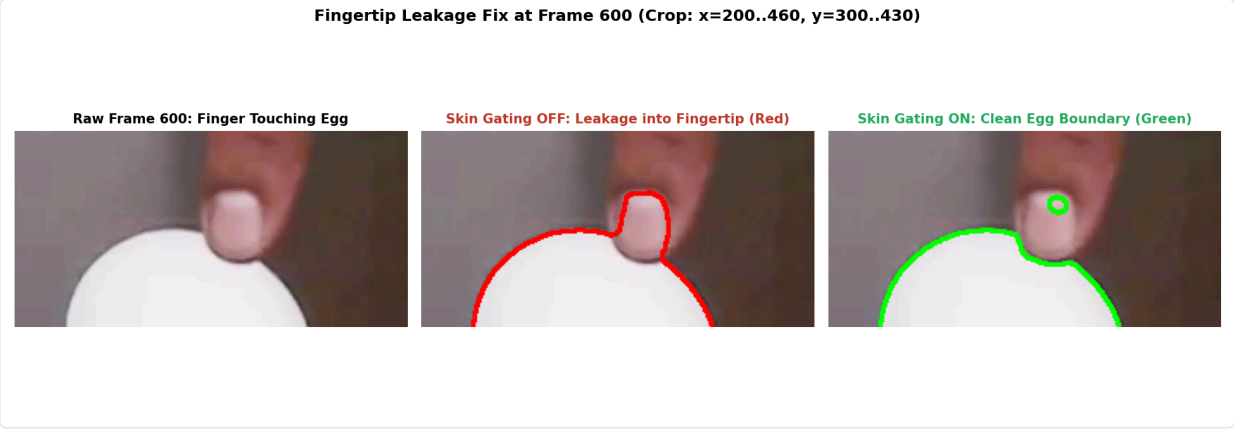


Figure 6: Frame 600 before/after comparison (crop: x ∈ [200, 460], y ∈ [300, 430]). Left: Raw frame. Middle: Baseline fingertip leakage. Right: YCrCb skin exclusion cleans egg boundary, eliminating 2296 px of skin noise.



Figure 7: 16-frame Video Tracking Contact Sheet tracking persistent identities through rotations and recoveries.

6. Limitations and Future Work

1. **Ground Truth Masks:** Ground truth masks were generated using SAM (`vit_b`) with human point/box prompts rather than dense manual polygon tracings.
2. **Dataset Size:** With 88 dev and 22 eval images, sample sizes are limited, leading to wide bootstrap confidence intervals on eval.
3. **Watershed Characteristics:** Distance transform watershed improved detection F1 on dev by separating touching eggs, but did not increase pixel IoU (it partitions instances without changing overall outer silhouette).
4. **Hand-Tuned Video Bounds:** Video tracking parameters (table ROI y ≥ 270, watermark bounds, skin thresholds) were calibrated for this specific video scene.
5. **Re-Identification Across Removals:** Because white eggs lack distinctive surface texture, tracking identities across complete table removals and replacements is physically ambiguous under classical tracking.
6. **Future Extensions:** Fine-tuning a deep semantic segmentation model (e.g. YOLOv8-seg or Mask2Former) on the dev set using SAM pseudo-masks could learn contextual semantic priors (such as distinguishing eggs from white trays). Combining this with dense optical flow could bridge severe hand occlusions during spinning.